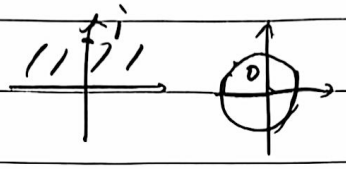
第七次作业

习题六 5(1) 6(2) 8 10 12 16(1)(3) 17

T5(1) $w=e^{i\theta}\frac{z-i}{z+i}$



$w'=e^{i\theta}\frac{z+i-z+i}{(z+i)^2}=e^{i\theta}\frac{2i}{(z+i)^2}$

$\arg w'|_{z=i}=\arg\frac{e^{i\theta}}{2i}=\arg e^{i(\theta-\frac{\pi}{2})}-\frac{\pi}{2}$ 故 $\theta=0$

$\therefore w=\frac{z-i}{z+i}$

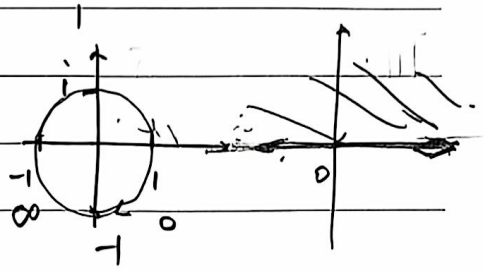
T6(2) $w=e^{i\theta}\frac{z-z_0}{\bar{z}z_0-1}$

$\Rightarrow w=e^{i\theta}\frac{z-\frac{1}{2}}{\frac{1}{2}z-1}$

$w(-1)=1 \Rightarrow e^{i\theta}=1 \Rightarrow \theta=0$

故 $w=\frac{2z-1}{z-2}$

T8 $w=K\frac{z-1}{z+1}$



$w|_{z=i}=K\frac{i-1}{i+1}=1 \Rightarrow K=-i$

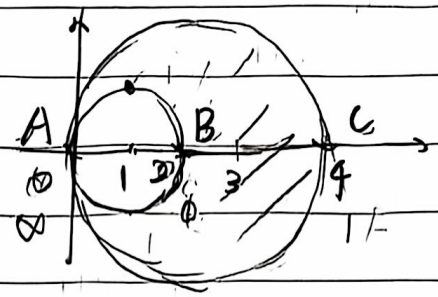
故 $w=-i\frac{z-1}{z+1}$

$w|_{z=-1}=-1$ 故此映射将 $\{z:|z|<1\}$ 映射为 $\{w=u+iv:v>0\}$

Date

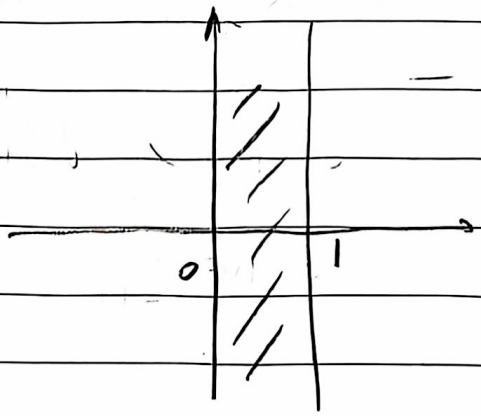
No.

T10 $w=\frac{z-i}{z+i}$



$w|_{z=1+i}=i$ $w|_{z=1-i}=-i$

故得A→B两条弧线映为y轴正负半轴



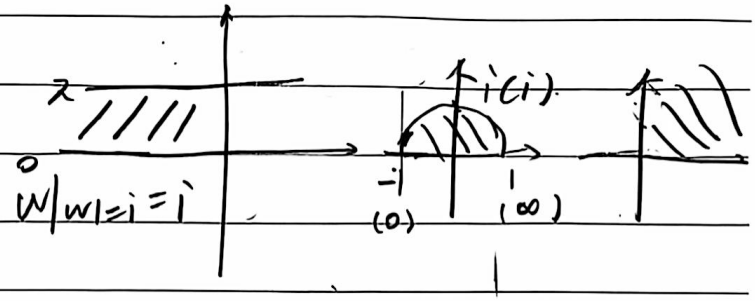
$w|_{z=2+2i}=1+i$ $w|_{z=2-2i}=1-i$

故得A→C两条弧线映为y=1正负半轴

即得区域A映为区域B

T12 $w_1=e^z$

$w=k\frac{w_1+1}{w_1-1}$



$w|_{w_1=i}=-ki$，取$k=1$，则$w|_{w_1=i}=i$

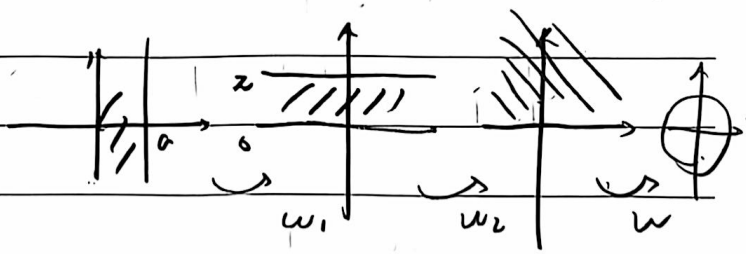
$w|_{w_1=0}=1$

故 $w=\frac{1+e^z}{1-e^z}$

T16 (1) $w_1=\frac{2\pi i}{a}z$

$w_2=e^{w_1}$

$w=\frac{w_2-1}{w_2+1}$



故 $w=\frac{e^{\frac{2\pi i}{a}z}-1}{e^{\frac{2\pi i}{a}z}+1}$

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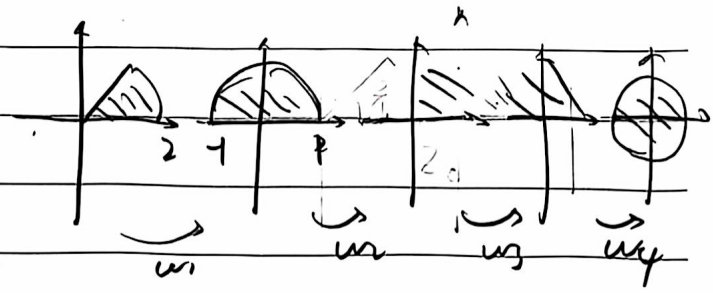
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(3) $w_1 = (\frac{1}{2}z)^4 = \frac{1}{16}z^4$

$w_2 = \frac{1+w_1}{1-w_1}$ $w_3 = w_2^2$

$w_4 = \frac{w_3-1}{w_3+1}$

故 $w = \frac{(\frac{16+z^4}{16-z^4})^2 - 1}{(\frac{16+z^4}{16-z^4})^2 + 1}$



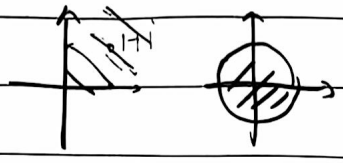
(1) $w_1 = z^2$

$z = 1+i$, $w_1 = 2i$; $z = 0$, $w_1 = 0$

故 $w = e^{i\theta}\frac{w_1 - 2i}{w_1 + 2i}$

故 $w_1 = 0$, $w = 1$ $\Rightarrow e^{i\theta} = 1$ $\theta = 2$

故 $w = -\frac{z^2 - 2i}{z^2 + 2i}$



$1+i \to 0$

$0 \to 1$

Data

No.

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